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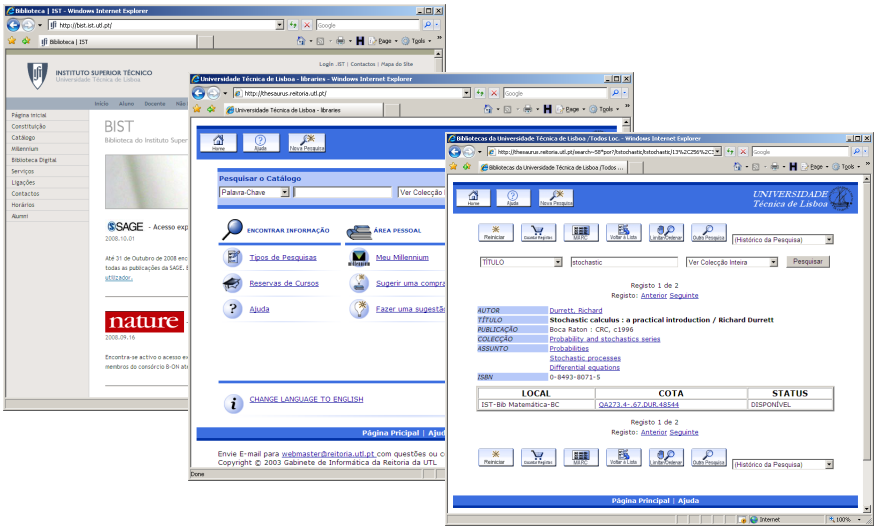

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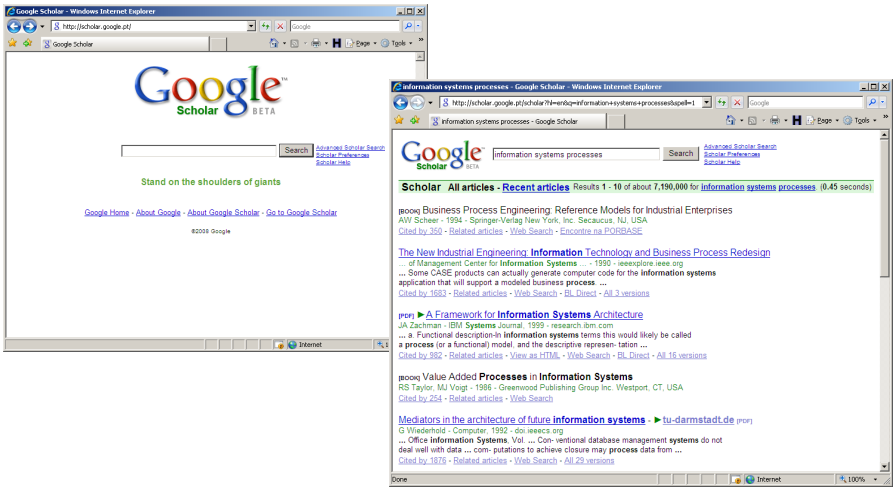

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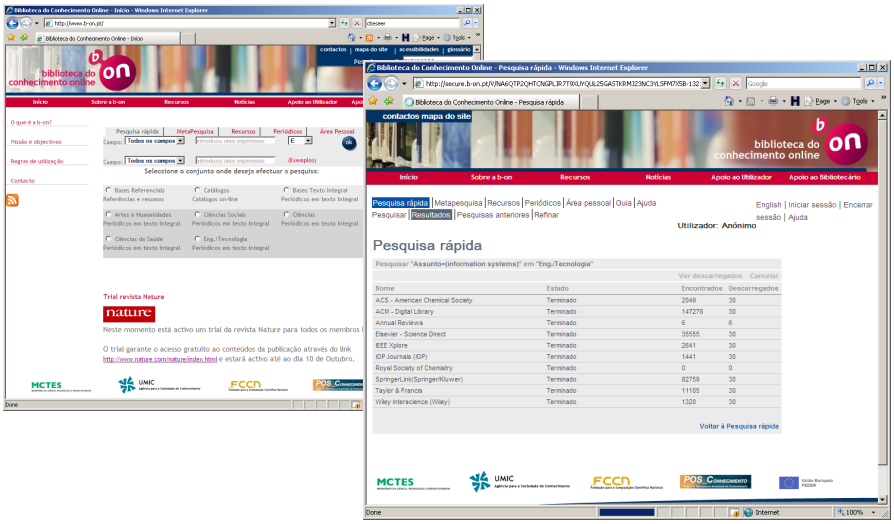

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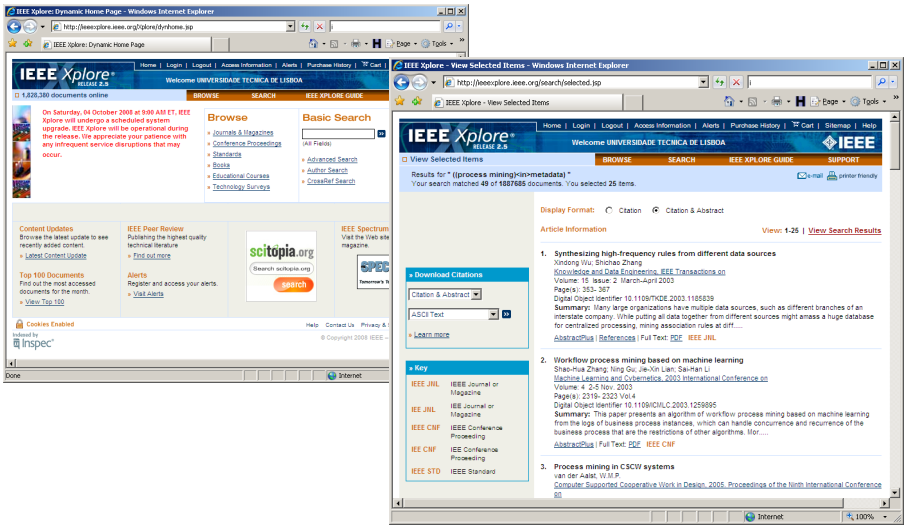

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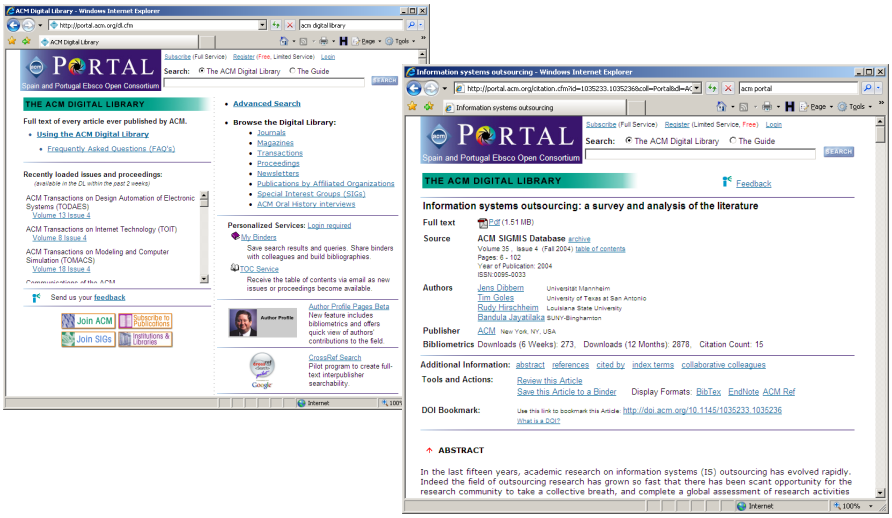

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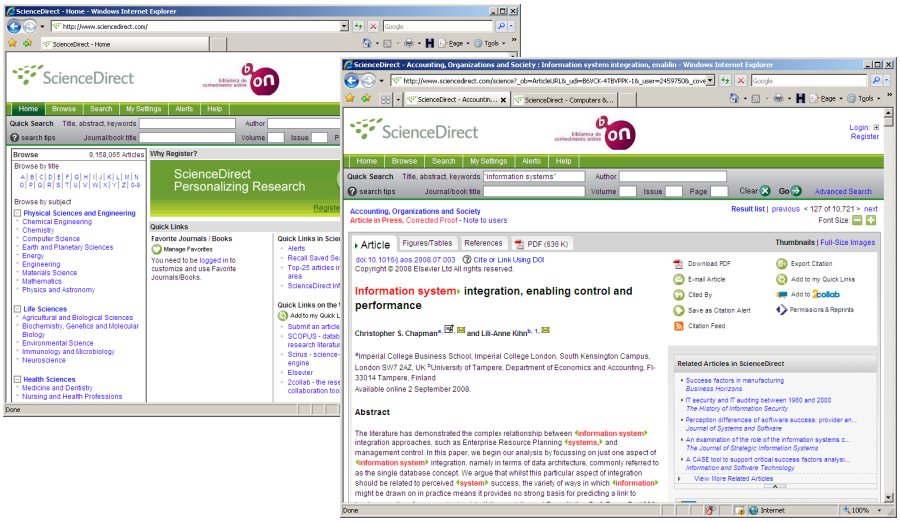

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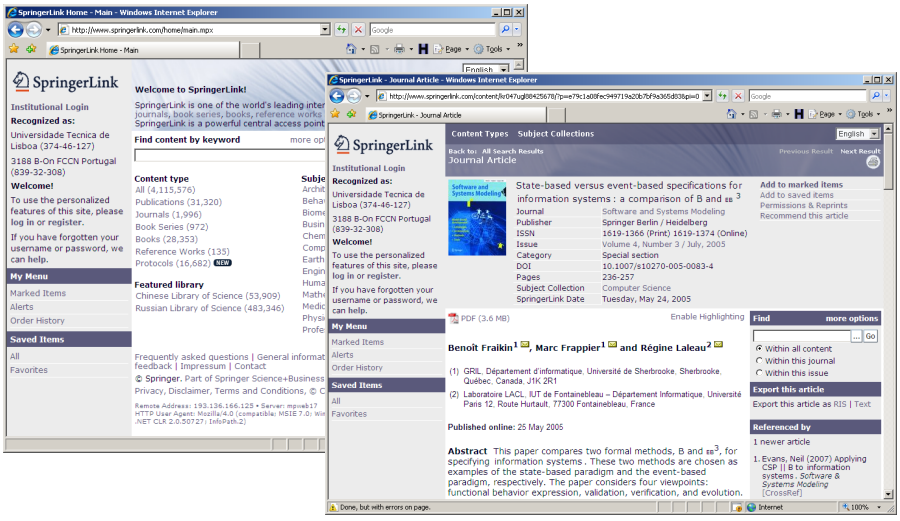

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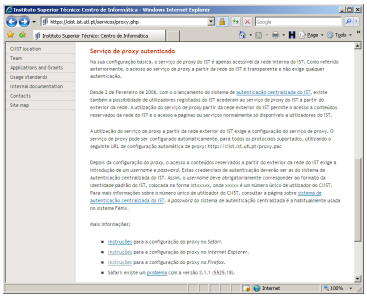

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On the Concurrency of Inter-organizational Business Processes

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Abstract. Within organizations, workflow systems can automate business processes by centrally coordinating activity sequences. But outside their borders, organizations are autonomous entities that cannot be subject to centralized process control. Their internal processes are autonomously defined and controlled, and what they need is to synchronize those concurrent processes. Just like Petri nets are a valuable tool to model activity sequencing in local business processes, π-calculus becomes a useful tool to model concurrency in inter-organizational processes. After a review of the main developments in cross-organizational workflow management, this paper illustrates the use of π-calculus to model the interactions between business processes running concurrently in different organizations. These interactions range from invoking external services to more complex patterns such as contract negotiation and partner search and selection. The paper concludes with a case study that illustrates the application of the proposed approach in a more realistic business scenario.

1 Background

Research on the application of workflow management in inter-organizational environments has produced a wealth of interesting approaches on how to deal with the problem of coordinating processes that span multiple organizations. The available solutions for cross-organizational workflow management have been developed in recent years and they have followed a path towards increasing flexibility, from focusing on "*low-level*" issues of workflow systems interoperability to more flexible, "*higher-level*" architectures based on contracts and views. Putting these developments in perspective, several trends can be identified, including:

Workflow interoperability mechanisms - after the publication of the Workflow Reference Model [1], which described four models of interoperability between workflow systems, supporting cross-organizational workflows seemed to be a matter of selecting the most appropriate interoperability model for a given scenario. In [2], Aalst̂ck and Dustdar describe the application of these models to medical imaging workflows that resemble cross-organizational workflows.

B. Mezeman, Z. Tari, P. Herrero et al. (Eds.): OTM Workshops 2008, LNCS 4277, pp. 844–852, 2008.
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In [3], van der Aalst formalizes the case-transfer and extended case-transfer models using Petri nets, and compares them as approaches to partitioning cross-organizational workflows over multiple business partners.

Federating heterogeneous workflow systems - it was quickly found out that run-time interoperability during process execution required build-time interoperability during process definition as well. The problem of connecting workflow systems then turned into a problem of federating them [4], i.e., to devise architectures in which several different workflow systems appear as a single, integrated one. Different solutions emerged, such as [5], in which Lindert and Deiters propose an approach to defining cross-organizational workflows via the interconnection of "process fragments" specified by different parties, in an early effort to address the problem of autonomy of business partners in inter-organizational settings. Reichert et al [6] realized the same problem, but focused on adaptive features that require the use of a centralized workflow modeling facility.

Agent- and service-based architectures - the paradigm of software agents renewed the interest in supporting cross-organizational workflows, especially in connection with service-oriented architectures, by facilitating the integration of local and remote services. Blake [7] developed a middleware architecture based on service-invoking agents which are essentially controlled by a global workflow manager agent that enforced a centralized workflow policy. Kwak et al [8] developed a middleware architecture where the workflow system inside an organization can invoke services registered either in a local service interface repository (LSIR) or in a global service interface repository (GSIR). Strickler et al [9] went even further to propose a trader system that promotes the reuse of workflow data types between organizations and supports bidding of service offerings to submitted service requests. Yan and Wang [10] devised an architecture where local processes are exposed as services that can be invoked in cross-organizational workflows.

Contract-based approaches - it had already been realized that the autonomy of business partners means that cross-organizational workflows are subject to agreements, or contracts. The CrossFlow project [11] was a milestone in the development of contract-based approaches, since it developed a consistent framework for establishing contracts and configuring workflow systems based on those contracts. Other authors have since then developed similar approaches, such as [12], in which van den Heuvel and Weigand propose a contract specification language to define cross-organizational workflows. But in [13], Kafeza et al realize that cross-organizational workflow contracts should refer only to partial views of local workflows. This is due to several reasons, including the need to keep some information private, or the fact that an organization may establish contracts with several partners having different requirements.

View-based models - the work on contract-based approaches opened up a new way of looking towards cross-organizational workflows. One of the most significant developments is [14], in which van der Aalst and Weske describe their public-to-private approach, where "public" stands for the agreed-upon workflow

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B initiates its chip-manufacturing process, while at the same time ordering the silicon wafers. Supplier C provides the wafers immediately and replenishes its own stock if needed. The wafers arrive in the middle stages of B's manufacturing process which, when completed, will provide the ASICs to company A.

6 Conclusion

Cross-organizational workflow management has been a long way from the run-time problem of supporting distributed workflow execution, up to the build-time problem of decoupling models of public collaborative workflows from those of private, local business processes. Still, it is clear that inter-organizational processes require solutions other than just mechanisms to enforce activity sequencing. The concurrent nature of inter-organizational processes calls for a solution that supports the interconnection of business processes running under the control of different organizations.

In this context, π-calculus becomes a valuable tool to model concurrency and synchronization in inter-organizational processes, just like Petri nets are useful to model activity sequencing in local business processes. In this paper, we have described mainly in a graphical way how these two formalisms can be combined in order to model inter-organizational systems. In the future, we will study the combination of these notations on formal grounds in order to determine the kind of properties that can be formally verified.

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Citações

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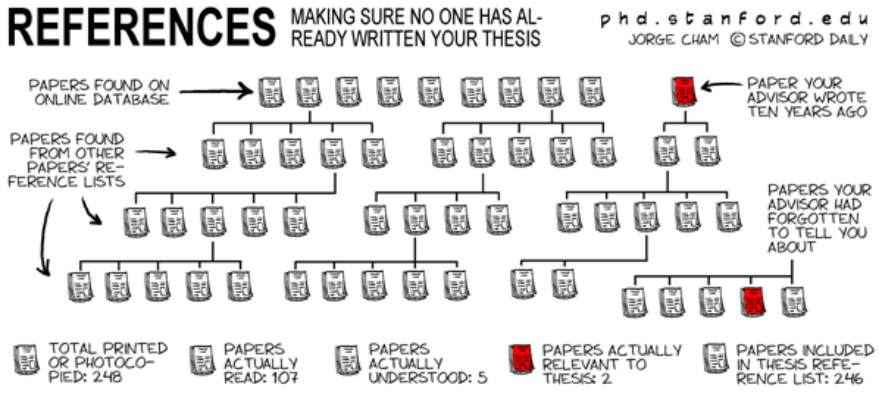
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